

The Solar Cycle

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Aim:

- Identify patterns in social trend data and geomagnetic data to determine the length of the solar cycle.
- Understand the link between sunspots, the geomagnetic field and auroral activity and why we use geomagnetic indices.
- Understand how social trends can reflect underlying natural cycles.

The sun's activity is responsible for auroras. This activity has a regular repeating cycle – your task is to figure out how long this cycle is. We will consider social trends data from Google Trends and the Trove archive, geomagnetic data and sunspot data and to do this.

Part 1: Google Trends data

Navigate to Google Trends: <https://trends.google.com/trends/>

What is Google Trends?

[Google Trends](#) is a free public tool that allows you to explore trends in Google searches for specific terms over time and in different areas of the world (Raffoul, 2024). It produces graphs and data tables that you can download and analyse. Data is presented on a scale of 0 – 100, where 100 is the peak interest in searches.

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1. Type 'aurora australis' into the search bar
 2. Change the location to 'Worldwide'
 3. Change the time to '2004 - present' (Google began collecting trends data in 2004)
 4. Leave the search type as 'Web Search'

Your search should look like **Figure A**.

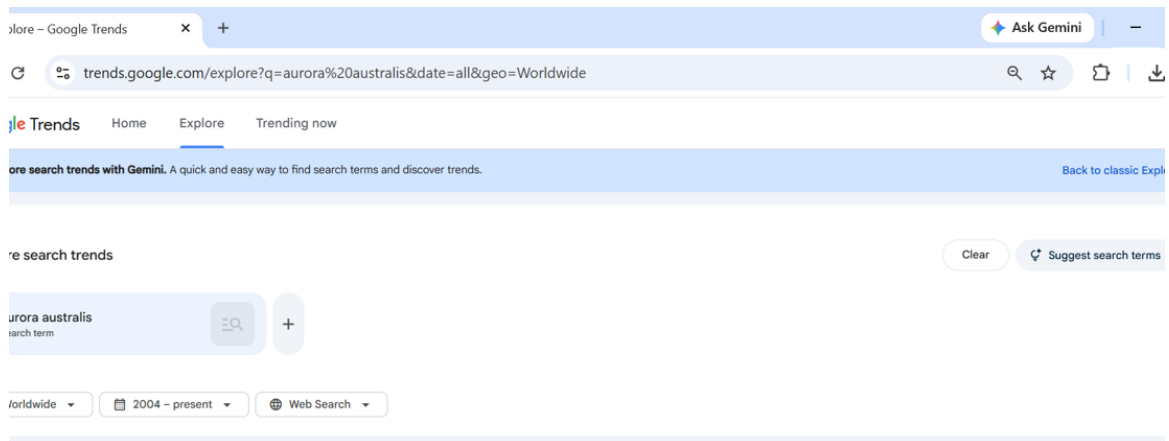


Figure A: Google Trends search for ‘aurora australis’ worldwide from 2004 – present (Google Trends, 2026).

A line graph will appear below your search criteria. Explore the graph by moving your mouse across the graph – you can click on points along the graph to see specific dates.

Your graph should look like **Figure 1** below – this figure is the same Google Trends data plotted using Python.

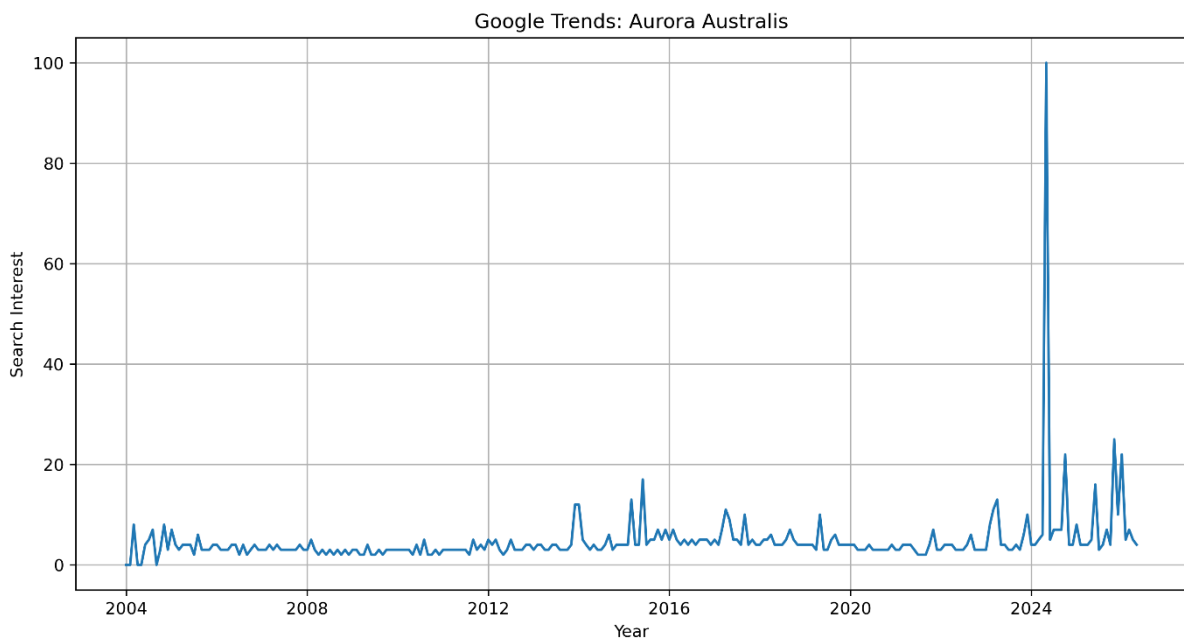


Figure 1: Worldwide search interest for ‘Aurora Australis’ (Jan 2004 – June 2026). Data source: Google Trends, 2026

(<https://trends.google.com/explore?q=aurora%2520australis&date=all&geo=Worldwide>)

Question 1: Why do you think we are searching for ‘aurora australis’ instead of just ‘aurora’ or ‘aurora borealis’?

Question 2: Is there a particular point on your graph where there is a big spike (i.e. peak interest) in Google searches for ‘aurora australis’? When was this? What do you think happened during this month?

Figure 2 below is the same plot but zoomed in to remove the big spike you just identified. This big spike is an outlier (a value far outside of the rest of the dataset) and makes trends in the rest of the data difficult to see.

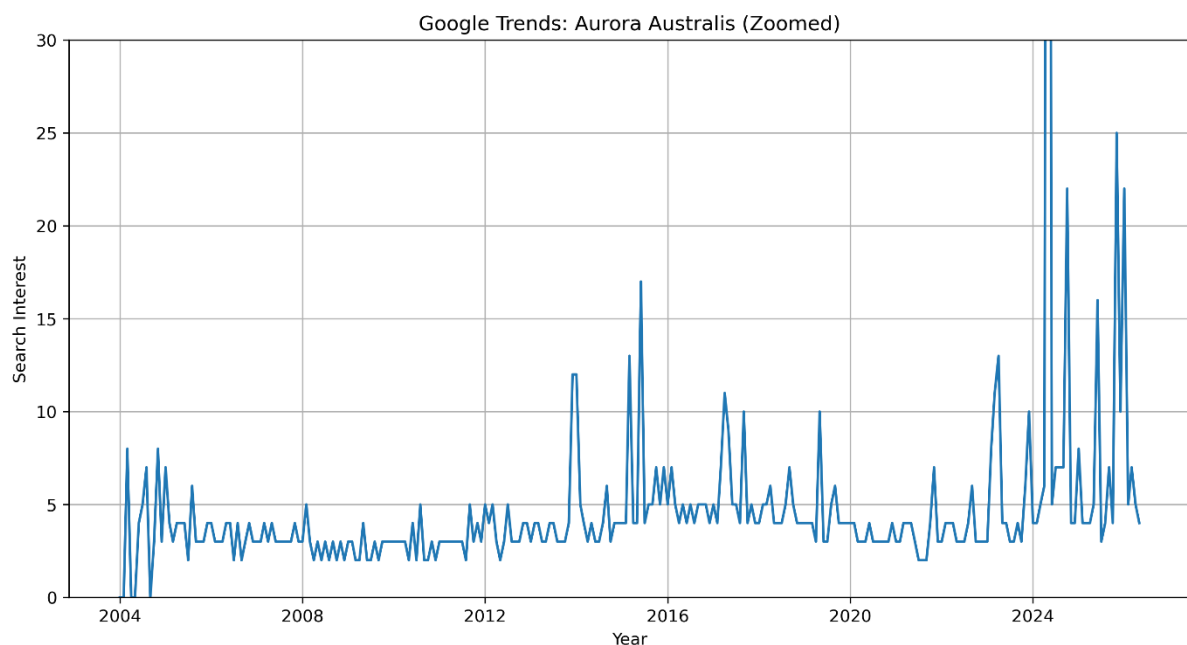


Figure 2: Zoomed in Worldwide search interest for ‘Aurora Australis’ (Jan 2004 – June 2026). Data source: Google Trends, 2026

(<https://trends.google.com/explore?q=aurora%2520australis&date=all&geo=Worldwide>)

Question 3: Can you identify any overall patterns on your graph i.e. search interest increasing to a peak and then decreasing back down? If you can, mark the boundaries on to your worksheet.

Do not worry if you cannot identify any convincing trends at this stage.

Is 2004-2026 long enough to see any trends? We will now consider social trends data from the Trove archive over a longer time period and see if that helps us.

Part 2: Trove archive data

Figure 3 below shows data from number of references to ‘Aurora Australis’ each year, according to the Trove archive of Australian newspapers.

The data is from 1822 to 2023 (Note that the Trove archive is limited by copyright restrictions – anything less than 70 years old is restricted for most sources, hence the counts drop away after 1956).

What is Trove?

[Trove](#) is an online platform from the National Library of Australia (NLA) and contains more than 14 billion digital items such as digitised newspapers, books, maps and photographs. Trove Content on the Trove database is fully searchable, making it an excellent research tool for all things Australian (ARDC, 2025; National Library of Australia).

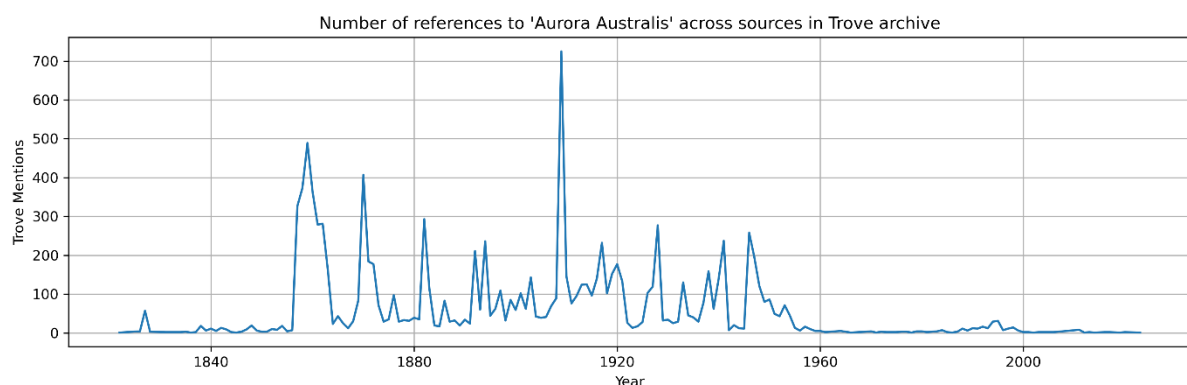


Figure 3: Number of references to “Aurora Australis” across sources in Trove archive (1822-2023), compiled from National Library of Australia Trove database.

Question 4: Can you identify any cyclical/ repeating trends in this data? If so, how long do cycles last? Mark the boundaries of any cycles you identify on your worksheet.

Hint: Try identifying peaks in Trove Mentions and measuring the amount of time between consecutive peaks?

Part 3: Geomagnetic observatory data

Geomagnetic observatories are permanent recording stations that measure changes in the Earth's magnetic field over time. There are geomagnetic observatories all around the world, and these are connected as a global network (INTERMAGNET).

Figure 4 is a graph of raw data from the Canberra geomagnetic observatory from 1991 – present. The data is for the x component, which shows geomagnetic activity in the magnetic north direction.

INSERT FIGURE – needs to be created by downloading and plotting Canberra geomagnetic x component data from 1991 – present from https://imag-data.bgs.ac.uk/GIN_V1/GINForms2

This could also be printed out as a large paper copy (multiple sheets taped together) and students could look at it from eye level with the table (similar to seismic interpretation) to try and see trends.

Figure 4: Geomagnetic activity (x component) from the Canberra Geomagnetic Observatory 1991 – the present. Data source: INTERMAGNET data portal

Question 5: Can you identify any patterns in this data over time? If not, why do you think patterns are hard to see?

Part 4: Indexed geomagnetic data

As you saw in part 3, the raw data collected at geomagnetic observatories (like the observatory in Canberra) is often complex and patterns are difficult to see.

Geoscientists use geomagnetic indices to solve this problem (Geoscience Australia). When indices are applied to raw data, noise is reduced and trends become easier to see.

In the field of geomagnetism, two important indices are the k index and the aa index.

- The k index measures geomagnetic activity every 3 hours relative to a quiet day on a scale of 0-9 for a specific observatory (Geoscience Australia).
- The aa index is an average measure of geomagnetic activity that uses the k index values from two observatories on opposite sides of the world – Canberra, Australia and Hartland, England (Geoscience Australia).

Figure 5 shows the yearly averaged aa index from 01 January 1868 to 01 June 2026. The monthly mean values are in blue, and the yearly average is in orange.

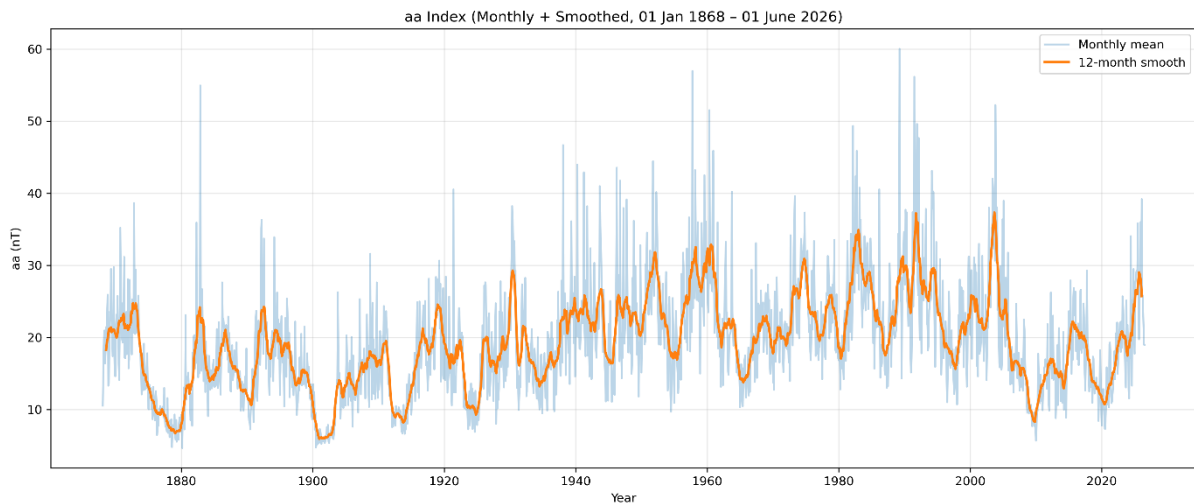


Figure 5: Geomagnetic aa index, monthly average (blue) and yearly smoothed (orange).
Data source: Mayaud, P.-N., Menvielle, M., & Chambodut, A. (2023). aa geomagnetic index. EOST. (Dataset). <https://doi.org/10.25577/9z05-v751>

Question 6: Do you notice any repeating patterns in the aa index data?

Question 7: Where are the peaks in geomagnetic activity? Label these and calculate approximately how many years there are between them.

Part 5: Sunspot numbers

We will now consider patterns in the number of sunspots over time.

Sunspots are dark, cooler regions on the surface of the Sun caused by intense magnetic activity (NASA, 2025). They are linked to solar eruptions, which drive geomagnetic activity on Earth. The number of sunspots varies over time.

Figure 6 shows how the number of sunspots has changed between 1770 to 2026.

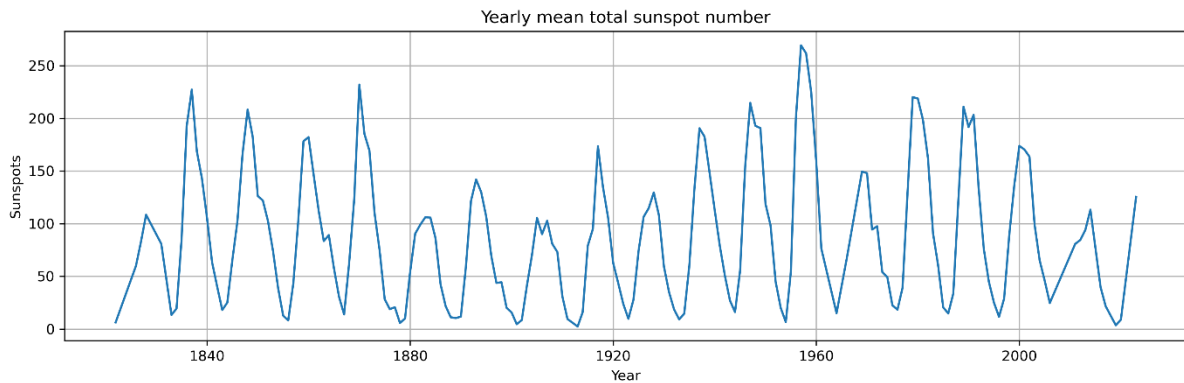


Figure 6: Yearly mean total sunspot number (1770-June 2026). Data source: WDC-SILSO, Royal Observatory of Belgium, Brussels, DOI: <https://doi.org/10.24414/qnza-ac80>

Question 8: Do you notice any repeating patterns in the sunspot data?

Question 9: Where are the peaks in sunspot number? Label these and calculate approximately how many years there are between them. These peaks are known as **solar maxima**.

Part 6: Summary

From your study of all the datasets, answer the following questions:

Question 10: You should have identified a regular repeating cycle in the Trove, aa and sunspot data. The length of this cycle should be approximately the same. How many years does one cycle take?

This is the duration of the **solar cycle**. The solar cycle is defined by a **solar maximum** in the middle of the cycle and a **solar minimum** at the cycle's start/ end (NASA, 2025).

Question 11: Is auroral activity stronger at the solar maximum or the solar minimum? Why do you think this?

The below figure (**Figure 7**) of the yearly mean total sunspot number and Trove archive 'aurora australis' mentions on top of one another may help you.

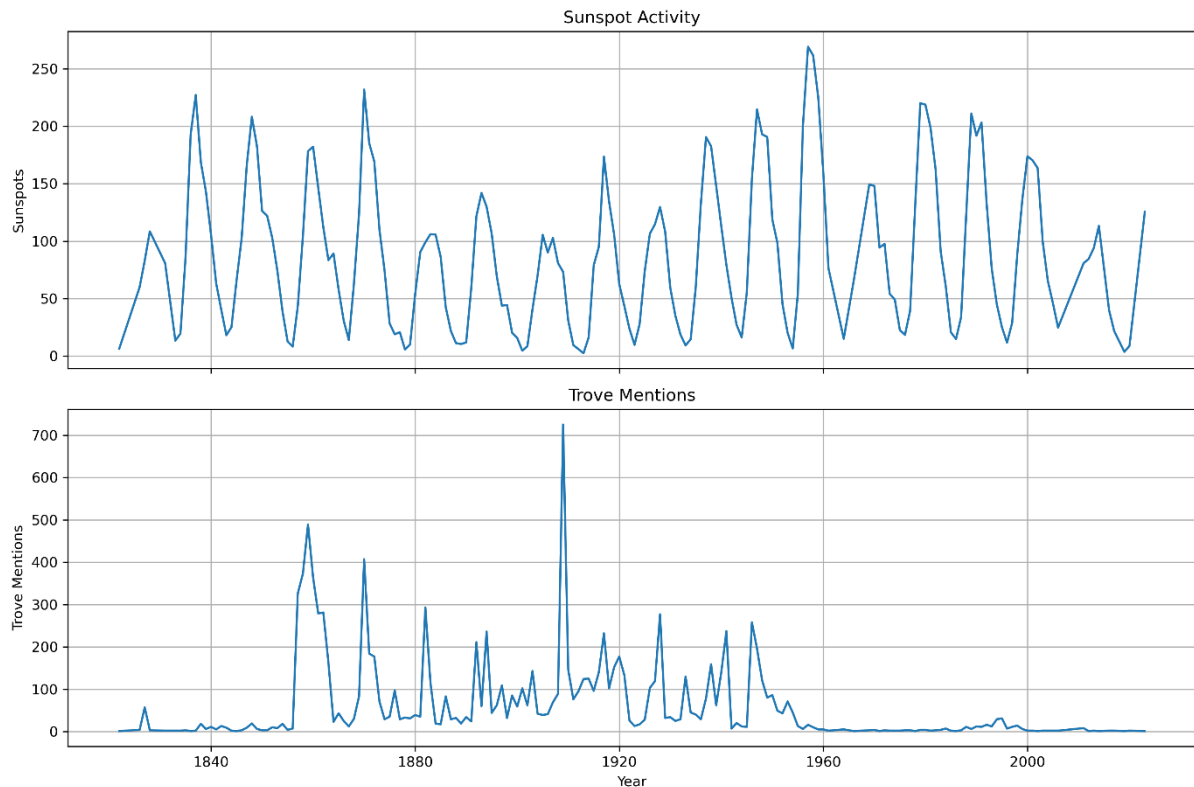


Figure 7: Yearly mean total sunspot number (1770-June 2026) (WDC-SILSO, Royal Observatory of Belgium, Brussels, DOI: <https://doi.org/10.24414/qnza-ac80>) above Number of references to “Aurora Australis” across sources in Trove archive (1822-2023), compiled from National Library of Australia Trove database.

Question 12: Why do you think the social trend data from Google Trends and the Trove archive closely resemble the solar cycle?

References:

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This resource was developed with assistance from ChatGPT (OpenAI, 2026), used to generate and refine instructional ideas.